

$$\left(n - \frac{2}{2}\right)^2 - \left(\frac{2}{2}\right)^2 + \left(\frac{2}{2}\right)^2$$

$$(n-1)^2 - 1$$

$$(i) \int \frac{8 \, du}{n^2 - 2n + 2}$$

This integral is solved by the completing the square in the denominator to transform it into the standard form $\int \frac{1}{u^2 + a^2} \, du$, which integrates to $\frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$.

so

$$8 \int \frac{dn}{n^2 - 2n + 2} = 8 \int \frac{du}{(n-1)^2 + 1}$$

$$(\text{let } u = (n-1))$$

$$a = 1$$

$$8 \int \frac{1}{1} \arctan\left(\frac{n-1}{1}\right) + C = 8 \arctan(n-1) + C$$

$$\text{Answer: } 8 \arctan(n-1) + C.$$

$$\int_0^{\infty} \frac{\ln y}{y+4y \ln^2 y} dy$$

~~$$\int \frac{\ln y}{y+4y \ln^2 y} dy$$~~

~~$$(x)$$~~

~~$$\text{let } u = \ln y$$~~
~~$$\frac{du}{dy} = \frac{1}{y}$$~~

$$= \int \frac{1}{y} \frac{\ln y}{1+4 \ln^2 y} dy$$

choose a substitution let $u = \ln y$.

$$\frac{du}{dy} = \frac{1}{y}$$

$$\int \frac{u}{1+4u^2} du$$

$$du = \frac{1}{y} dy$$

performing the second substitution

$$\text{let } v = 1+4u^2$$

$$\frac{dv}{du} = (0+8u)$$

$$dv = (0+8u) du =$$

$$\frac{1}{8} dv = u du$$

Now Re-write the integral in terms of v .

$$\int \frac{1}{v} \left(\frac{1}{8} \right) = \frac{1}{8} \int \frac{1}{v} dv = \frac{1}{8} \ln |v| + c$$

substitute back $v = 1+4u^2$

$$\frac{1}{8} \ln|1+4u^2| + c$$

iii/ $\int n e^{3n} dn$

ILATE

let $u = n$

$$v = e^{3n}$$

formulae-

$$\int u \cdot v = u \int v du - \int \frac{du}{dn} \cdot \int v du.$$

$$n \left[\frac{1}{3} e^{3n} - \left[\int 1 \cdot \frac{1}{3} e^{3n} \right] \right]$$

$$n \left(\frac{1}{3} e^{3n} \right) - \frac{1}{3} \int \frac{1}{3} e^{3n} dn$$

$$n \left(\frac{1}{3} e^{3n} \right) - \frac{1}{3} \left[\frac{1}{3} e^{3n} \right] + c$$

$$n \frac{1}{3} e^{3n} - \frac{1}{9} e^{3n} + c$$

$$\frac{1}{3} n e^{3n} - \frac{1}{9} e^{3n} + c \text{ Ans.}$$

iv, $\int \tan^4 x \sec^4 x \, dx.$
 sol

$$\int \tan^4 x \cdot \sec^2 x \cdot \sec^2 x \, dx.$$

Use the identity $\sec^2 x = 1 + \tan^2 x.$

$$\int \tan^4 x (1 + \tan^2 x) \sec^2 x \, dx$$

Let $u = \tan x$

$$\frac{du}{dx} = \sec^2 x.$$

$$du = \sec^2 x \, dx.$$

$$\int u^4 (1 + u^2) \, du$$

⑥ Distribute and integrate-

$$\int (u^4 + u^6) \, du = \frac{u^5}{5} + \frac{u^7}{7} + c$$

7 substitute back $u = \tan x.$

$$\frac{\tan^5 x}{5} + \frac{\tan^7 x}{7} + c$$

Ans

Now factor the Numerator

$$1-u^2 = (1-u)(1+u).$$

$$\int \frac{(1-u)(1+u)}{\sqrt{1-u}} du.$$

Now simplifying fraction using the property.

$$\frac{a}{\sqrt{a}} = \sqrt{a}$$

$$\int \sqrt{1-u} (1+u) du$$

$$\int (1-u)^{\frac{1}{2}} (1+u) du.$$

Now perform another substitution

$$(1-u)^{\frac{1}{2}}: \text{ let } w = 1-u$$

• Find the differential $dw: dw = -du = -du$.

Express $(1+u)$ in terms of w . So

$$w = 1-u, \text{ then } u = 1-w.$$

Rewrite integral in terms of w .

$$\int w^{\frac{1}{2}} (2-w) (-dw) = -\int (2w^{\frac{1}{2}} - w^{\frac{3}{2}}) dw$$

$$-\left[2 \frac{w^{\frac{3}{2}}}{\frac{3}{2}} - \frac{w^{\frac{5}{2}}}{\frac{5}{2}} \right] + C = -\left[\frac{4}{3} w^{\frac{3}{2}} - \frac{2}{5} w^{\frac{5}{2}} \right] + C$$

$$\frac{2}{5} w^{\frac{5}{2}} - \frac{4}{3} w^{\frac{3}{2}} + C.$$

$$\Rightarrow \text{ii) } \int \cos^4 x \, dx$$

$$\int \left(\frac{1 - \cos 2x}{2} \right) \left(\frac{1 + \cos 2x}{2} \right)^2 dx$$

$$\frac{1}{8} \int (1 - \cos 2x) (1 + \cos 2x)^2$$

$$\frac{1}{8} \int (1 - \cos 2x) (1 + 2\cos 2x + \cos^2 2x) dx$$

$$\frac{1}{8} \int 1 \times (1 + 2\cos 2x + \cos^2 2x)$$

$$= 1 + 2\cos 2x + \cos^2 2x$$

$$\text{Now } -\cos 2x (1 + \cos^2 2x + \cos^2 2x)$$

$$-\cos 2x - 2\cos^3 2x - \cos^3 2x$$

And result.

$$1 + \cos 2x - \cos^3 2x - \cos^3 2x$$

$$= \frac{1}{8} \int (1 + \cos 2x - \cos^3 2x - \cos^3 2x) dx$$

$$\frac{1}{8} \left[x + \frac{1}{2} \sin 2x - \int (\cos^5 2x - \cos^3 2x) dx \right]$$

For the term involving $\cos^2 2x$, we

use.

$$\int \cos^2 2x \, dx = \frac{1}{2} \int (1 + \cos 4x) dx$$

$$= \frac{1}{2} \left(\frac{x}{1} + \frac{\sin 4x}{4} \right)$$

For $\cos^3 2n$ term, we have

$$\int \cos^3 2n \, dn = \int (1 - \sin^2 2n) \cos 2n \, dn$$

$$\text{let } u = \sin 2n$$

$$\frac{du}{dn} = \cos 2n$$

$$du = 2 \cos 2n \, dn$$

$$\frac{1}{2} \int (1 - \sin^2 2n) \cos 2n \, dn$$

$$\frac{1}{2} \int (1 - u^2) \, du = \frac{1}{2} \left(u \sin 2n - \frac{1}{3} \sin^3 2n \right)$$

Now combining

$$\sin^2 n \cos^4 n \, dn = \frac{1}{16} \left(\frac{n-1}{4} \sin^4 n + \frac{1}{3} \sin^3 2n \right)$$

$$\text{vii) } \int \frac{-2n+4}{(n^2+1)(n-1)^2} dn$$

$$\int \frac{An+B}{(n^2+1)} + \frac{C}{(n-1)} + \frac{D}{(n-1)^2}$$

→ Multiply $(n^2+1)(n-1)^2$

$$(An+B)(n-1)^2 + C(n^2+1)(n-1) + D(n^2+1)$$